Trigonometry in 3D

1

a

i $EG = 7\sqrt{2}$ cm ii $AG = 7\sqrt{3}$ cm

b

i 45° ii 35° iii 90° iv 55°

2

a $z = 14.32$ cm **b** $\theta = 11.83^\circ$

3

a $z = 21.47$ cm **b** 75.15 cm^2

4

a 11.31° **b** $CE = \sqrt{325}$ **c** $BE = \sqrt{334}$ **d** 9.45°

e $CX = \sqrt{250}$ **f** $z = 10.74^\circ$

5

a $x = 15$ **b** $y = 17$ **c** $\theta = 28^\circ$

6

a

i $\alpha = 90^\circ$ ii $\beta = 90^\circ$ iii $\gamma = 90^\circ$

b $x = \sqrt{2}$

c $y = \sqrt{3}$

d $\theta = 35^\circ$

7

a $b = \sqrt{746}$ **b** 109 cm^2 **c** $\theta = 16^\circ$

8

a 91.92 m **b** 37.14 m **c** $11^\circ 25'$ **d** $15^\circ 57'$

9

a $BC = 82$ m **b** 48 m

10 $\theta = 65^\circ 9'$

11

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12

a $MD = \sqrt{32}$ cm **b** $x = 61.87^\circ$ **c** $h = \sqrt{112}$ cm **d** $MN = 4$ cm

e $PN = \sqrt{128}$ cm **f** 69.30° **g** 70.53°

13 $AF = 55.77$

14

a $h = 9.37$ m **b** $\theta = 25^\circ$

15

a $x = 56$ **b** $h = 90$

16

a $h = 47$ **b** $x = 131$ **c** $y = 97$

17

a $\overline{AE} = 26.0$ m **b** $\theta = 31^\circ 14'$

18

a 2481 m **b** 1486 m **c** 1384 m

19

a

i $VA = 8.49$ cm **ii** $VC = 7.81$ cm

iii $AC = 7.81$ cm

b $C = 65^\circ 51'$

20

a $CD = 22.65$ m **b** $AB = 18.29$ m **c** 16.18 m

21 41 m

22

a $d = \sqrt{41}$ m **b** $h = 9.86$ m **c** $x = 63.11^\circ$ **d** $y = 67.92^\circ$

23

a 1282 m **b** 2033 m **c** $AB = 1977$ m

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